MLR - Categorical Variables

Hello and welcome to another session on Business Statistics for Entrepreneurs. In the last module we discussed multiple linear regression, where a dependent variable y was a function of multiple independent variables x1, x2 and so on up to xp. In this module, we are going to continue our journey of statistical inference and look for an important extension of multiple linear regression, where this dependent variable y will be permitted to be a function of multiple independent variables x1 through xp, where some of them could be categorical or qualitative variables.

Joining me in this journey are our bright students, Tejas and Ashwini. Along with them, I will cover some important concepts and relevant applications. As always, they are going to keep me honest and on track. Welcome back Tejas and Ashwini. Are you ready for multiple linear regressions with categorical variables?

Yes, the example of flat prices in Mysore is still bothering me. I would hope that adding categorical variables help us get a better multiple linear regression model. Yes, in many of the data sets we have seen so far, there are many qualitative variables. It will be good to know how to use them in our regression models.

Yes, and our analysis will follow along very similar lines as what we have seen in the last module. So, what is our plan for this module? Let me start with an overview of what we are going to be covering in this module. We will start with the basic idea behind multiple linear regression with categorical variables.

As before, we will try to establish a linear relationship between a dependent variable y and several independent variables x. However, now some of these independent variables can be qualitative or categorical variables. The other variables could be quantitative, like we had in the previous module. We start with the basic multiple regression model and the associated equation.

We will also see the estimated multiple regression equation. And as we did in the previous module, the least square technique will be used to determine the coefficients of the multiple

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regression equation. We will evaluate our model based on the multiple coefficient of determination and like we did in the previous module, we will have the test for overall significance, namely the F test and the test for individual significance, namely the T test.

We will analyze the residuals as we have always done. And of course, we will apply them to examples, some of the same examples that we have covered earlier, just to see how the inclusion of these categorical variables impacts the performance of the overall model. Let us get started. What is the basic idea behind multiple linear regression with categorical variables?

Like in multiple linear regression, we try to establish a linear relationship between a dependent variable y and several independent variables x. However, now we want that some of these p independent variables can be qualitative or categorical variables. The other variables could be quantitative like we had in the previous module.

This is the essential difference between multiple linear regression that we saw in the previous module and the multiple linear regression model with categorical variables that we are going to see in this module. This simple, yet important extension allows us to consider more factors that impact the dependent variable.

And this allows us to obtain better predictions than what would be possible with multiple linear regression without categorical variables. And Ashwini and Tejas, you will agree that in some of the examples we did in the last module, it did appear that some of the categorical variables, if included in our multiple linear regression model, could have given us far better predictions.

In fact, in the examples that we have covered, they have involved only quantitative independent variables. For example, in Vijay Gowda's data set with flat prices in Mysuru, we looked at quantitative variables such as the area square feet, the number of bathrooms, and the number of bedrooms. We could not consider the effect of qualitative variables, such as the nature of the apartment community or complex, the amenities available, and whether the apartment was located in a premium location.

Yes, and in Manjula Nayak's data set, we consider only first income, monthly payment and utilities. Yes, and for that household survey in Bengaluru, we considered only the quantitative

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variables such as the first income, the monthly payment and utilities. We could not include the qualitative parameters such as the location of the house or if the household is owning or renting their house.

In fact, when we looked at Basavaraja's data on customer satisfaction score, we had only a single quantitative variable price. Yes, that is true, Tejas. And when we built a simple linear regression model using that data set, to develop a simple linear regression equation, we used the quantitative variable of price as the independent variable.

We had other qualitative variables also in our data sets, such as the brand of the computer, the attributes of the computer, such as RAM and the hard disk, and the operating system. We were unable to include any of such information in our regression models. Now, we will see how to do that. As you can see, in many situations, we must work with categorical independent variables.

Our purpose is to see how such categorical variables are handled in multiple linear regression analysis. It is probably best and easiest to understand the use and the interpretation of categorical independent variables by considering one of our examples that we have looked at earlier in the context of simple linear regression.

Let us start with Basaraja's dataset that had only one independent variable that was quantitative. And the others were all qualitative. Let us revisit that example and remind ourselves of what we got for simple linear regression in that case. If you recall, the data set had 36 responses from a survey.

It had variables like score, the price of a computer, the brand, and features like RAM, hard disk, etc. Basaraja was interested in knowing what factors might be influencing the satisfaction score given by customers. As part of simple linear regression, we analyzed if price is a significant determinant of the satisfaction score.

Score was our dependent variable and price was our independent variable x. The estimated regression equation was b 0 plus b 1 times price. And what was that given by? Do you remember? The equation we got was that Y hat is equal to 48. 44 plus 0. 132 times X. Yes.

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And what was the R square value? That was only 0. 36. Yes. And that was suggesting that there may be other factors other than price that is causing the variation in the scores. The F statistic in our F test was the ratio of MSR over MSC. And that came out to be 18. 9. And the corresponding p value was close to zero. And therefore, we conclude that we can reject the null hypothesis and claim that a linear relationship exists between score and price, and that relationship is significant.

When we did the t test, we got the test statistic to be equal to 4. 36. For this t statistic, the p value was close to zero, again confirming that the linear relationship between price x and the score y is significant. Now, let us say that we continue to maintain the satisfaction score as the dependent variable y.

And, since we have already established that price has a significant linear association with the satisfaction score, let us retain that also to be one of the independent variables. Now, let us say that we believe that a satisfaction score is related to the brand of the computer being purchased by the customer.

In the dataset, we have information that the customers who responded to the survey either purchased a Lenovo brand or a Dell brand of computer. And we know that we have data for 36 respondents of the survey. Let Y denote the customer satisfaction score and x1 denote the price paid for the computer.

Then, the regression model that uses only price as x1 to predict y, the satisfaction score, is given by y is equal to beta 0 plus beta 1 times x 1 plus the error term epsilon. Using the analysis tool pack, we found that the estimated regression equation was given by y hat is equal to 48. 43 plus 1. 32 times the price in lakhs.

To incorporate the brand of the computer into the regression model, we now define the following variable x2. We define that x2 is equal to 1 if the brand is Lenovo, and x2 is equal to 0 if the brand is not Lenovo, that is, if it is a Dell computer. In the regression analysis, x2 is called as a dummy or an indicator variable.

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Using this dummy variable, we can write the multiple linear regression model as y is equal to beta 0 plus beta 1 times x1 plus beta 2 times x2 plus the error term epsilon. You can see the revised data set here that includes the value of the dummy variable. Note that the dummy variables can take values of 0 or 1 only.

This implies that if you have two brands as we did in this case, we could denote one of the brands, Lenovo in our example, by the variable x2 taking a value of 1. And then, the value of x2 corresponding to 0 would denote the other brand, namely Dell in our case. Let us now analyze the multiple linear regression model that we have, with x1 corresponding to price and x2 corresponding to the dummy variable brand.

That takes a value of 1 if the brand is Lenovo and 0 if the brand is a Dell computer. Let us use the analysis tool pack to determine the estimated regression equation in this case. We go to analysis tool pack and select the Y variable to be the satisfaction score and the X variables to be the columns corresponding to price as well as the dummy variable corresponding to whether the brand is Lenovo or not.

As we have done always, let us collect not only the numerical outputs, but also the data set of standardized residuals and the plot of the residuals. Let me wait until you have collected all of those outputs before we begin to discuss. Yes, I have all the outputs now. Okay, so now let us discuss what we see in the model.

What can you say about the R square for this model? The R square of the model has increased to 0. 45 from the 0. 31 that we had. Yes, and when we had only price in the model, so it seems that that is a pretty good improvement. Adding the dummy variable corresponding to the brand does seem to have improved the model quite a bit.

And what about the standard error? The standard error in the model is now 4. 85. Yes, and that has reduced from the earlier standard error of 5. 19. Let us now look at outputs of ANOVA analysis. The table shows that the sum of squares regression and the sum of squares error are 648. 2 and 777. 4. Yes, and in the earlier regression model, the sum of squares regression, SSR, was 510.

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8. That has now improved to 648. 3. In the earlier model, the sum of squares error was 914. 8 and that has now reduced to 777. 4. Note that the sum of squares total SST is the same in both models and is equal to 1425.6.

The computation of MSE and MSR follow the same principles as we saw earlier for multiple linear regressions. So, what is our F statistic in this case? It is 13. 75 given a p value close to 0. Yes. And so that implies that we can reject the null hypothesis that both beta 1 and beta 2 are equal to 0. And the linear relationship between price, brand, and satisfaction score is significant.

Let us now look at the table with the coefficients. What are the values of B0, B1, and B2? B0 is 46. 83, B1 is 1. 29, and B2 is 3. 93. Yes, and what does our t test for individual significance say in this case? In this case, we look at the t test corresponding to the price in lakhs first. We find that the standard error is 0. 283, and that the coefficient of B1 is equal to 1. 29. To get the t statistic, we divide this coefficient, which is 1. 29, divided by the standard error, which is 0. 283, and that gives us a t statistic value of 4. 58. Now, this test statistic is a t random variable with 33 degrees of freedom. We find the p value corresponding to this test statistic, and what is that value, Ashwini?

We see that it is fairly close to zero. And that allows us to reject the null hypothesis that beta one is equal to zero. And this implies that in the population, price has a significant linear association with the satisfaction score provided the customers. Next, let us look at the t test corresponding to the brand. We find that the standard error is 1.63 and the coefficient B2 is equal to 3. 93. To get the test statistic, we divide the coefficient which is 3. 93 by the standard error which is 1. 63. And what do we get? We get a t statistic value of 2. 42. Yes, this is the test statistic and it is also a t random variable with 33 degrees of freedom. We find the p value corresponding to this test statistic and we see that it is fairly close to zero.

That allows us to reject the null hypothesis that beta 2 is equal to zero. So this implies that in the population brand has a significant linear association with the satisfaction score provided by the customers.

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Interpretation of Coefficients

Let us now talk about how we interpret the parameters of beta1 and beta2 when you have categorical variables in multiple linear regression. The multiple linear regression equation for Basaraju's data set at Chroma is given by the expected value of y, E of y, is equal to beta 0 plus beta 1 times x1 plus beta 2 times x2.

How do we interpret the parameters beta 0, beta 1, and beta 2 when a categorical variable is present? Let us first consider the case where x2 is equal to zero. This corresponds to the brand being Dell. Let us denote the expected value of y given brand is equal to Dell by E of y given Dell. Then, the E of y given that the brand is Dell is given by beta 0 plus beta 1 times x1 plus beta 2 times 0, which is equal to beta 0 plus beta 1 times x1.

Similarly, when x2 is equal to 1, this corresponds to the brand being Lenovo. Let us denote the expected value of y given that the brand is equal to Lenovo by E of Y given Lenovo. Then E of Y given that the brand is Lenovo is given by beta 0 plus beta 1 times x1 plus beta 2 times x2, which is equal to beta 0 plus beta 2 plus beta 1 times x1. Comparing these equations, we see that we have the average satisfaction score is a linear function of x1 for both the brands Dell and Lenovo. The slope of both these equations is beta 1, but the y intercepts are differ. What is the y intercept corresponding to Dell? The y intercept is beta 0 in the equation corresponding to the brand being equal to Dell.

Yes, and what about the y intercept for Lenovo? And the y intercept is equal to beta 0 plus beta 2 in the equation. Corresponding to the brand being equal to Lenovo. So the interpretation of Beta 2, therefore, is that it indicates the difference between the satisfaction score for the Lenovo computer compared to the Dell computer.

If Beta 2 is positive, then the average satisfaction score for Lenovo computer will be greater than the average satisfaction score for a Dell computer. If Beta two is negative, then the average satisfaction score for a Lenovo computer will be less than the average satisfaction score for a Dell computer.

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And what if beta 2 is equals to 0? Then there is no difference in the average satisfaction score between a Lenovo computer and a Dell computer. And therefore, the satisfaction score is not related to the brand. Using the estimated regression equation, y hat is equal to 46. 84 plus 1. 29 times the price, or x1, plus 3. 93 times x2. We see that 46. 84 is the estimate of beta 0, and 3. 93 is the estimate of beta 2. So, when x2 is equal to 0, that is when the brand is Dell, y hat is given by 46. 83 plus 1. 29 times x1.

And when x2 is equal to 1, the corresponding to the brand being Lenovo, we see that y hat is equal to 46. 83 plus 1. 29 times x1 plus 3. 93 times 1, which comes out to be equal to 50. 77 plus 1. 29 times x1. What this does is that by using a dummy variable for the brand of computer, we get two estimated regression equations that can be used to predict the satisfaction score. One corresponding to the brand being Dell and the other corresponding to the brand being Lenovo.

Exactly and in addition with B2 being equal to 3. 93 we learn that on an average the Lenovo computer results in a satisfaction score that is 3. 93, more than the satisfaction score of a Dell computer. In this figure, you can see the plots of the estimated regression equation for both the Lenovo computer and the Dell computer. This is pretty nice being able to incorporate the effect of brand in this way.

Yes, it is. Okay. Now before we go further, what do we see with the residuals? Let us do our usual check of the residuals. The plot of residuals against price x1 seems to be distributed above and below the x axis. Yes, and notice that the plot of residuals against x2, which takes values of 0 and 1, our categorical variable, is also above and below the x axis, but now clustered at the two values that x2 can take.

Looking at the standardized residuals, there are couple values above and below 2. Yes, and the plot of residuals against y hat also looks okay. And so does the plot of the standardized residuals against y hat. But, what if we had three brands, let's say Lenovo, Dell, and Asus? Yes, that's a good question. How would you model it if there were three brands?

Have the dummy variable take three values 0, 1, and 2 corresponding to the three brands? Well, no, we don't do it that way. In fact, what we do is to use two dummy variables that each takes

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values of 0 or 1. Let us look at this in more detail. If a categorical variable has k levels, then k minus 1 dummy variables are used and each dummy variable being coded as zero or one. Oh.

How does that work? Well, let us take a hypothetical case where we are trying to develop a regression model for satisfaction score, y, only as a function of the brand. So you mean ignore price for now. Yes, just to make our discussion a little bit easier, and let us say that we had three brands of computers, Lenovo, Dell, and Asus.

Then we would use a regression model that has two dummy variables, x1 and x2, to model the effect of the presence of these three brands. Our regression equation for the expected satisfaction score would be written as expectation of y or E of y is equal to beta 0 plus beta 1 times x1 plus beta 2 times x2, where we will first define the first dummy variable x1 to be equal to 1 if the brand is Lenovo and 0 otherwise.

And we will define x2 to be equal to 1 if the brand is Dell and 0 otherwise. With this definition, we would have the following values of x2 and x3 for the three brands. If the brand is Lenovo, then x1 would be 1 and x2 will be 0. If the brand is Dell, then x1 would be 0 and x2 would be 1. And if the brand is Asus, then both x1 and x2 will be 0.

The regression equation, then relating the expected value of the satisfaction score, E of y, to the dummy variables would then be written as E of y is equal to beta 0 plus beta 1 times x1 plus beta 2 times x2. Then, we would have three variations of this regression equation. The expected value of y, given that the brand is Asus, that is, E of y given Asus, will be equal to beta 0 plus beta 1 times 0 plus beta 2 times 0, which is simply equal to beta 0.

Expected value of y, given that the brand is Lenovo, that is E of Y given Lenovo is equal to beta 0 plus beta 1 times 1 plus beta 2 times 0, which is equal to beta 0 plus beta 1. The expected value of Y given that the brand is Dell, that is E of Y given Dell, will be equal to beta 0 plus beta 1 times 0 plus beta 2 times 1 which is equal to beta zero plus beta two. What would be the interpretation of beta zero? Yes. So in this case, beta zero will be the expected satisfaction score for the Asus brand of the computer. Beta one will be the difference between the expected satisfaction score between the Asus brand and the Lenovo brand.

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And beta two will be the difference between the expected satisfaction score for the Dell brand and thesuCEs brand. The two dummy variables were required because the brand in this case was a categorical variable with three levels. How do you decide which Exis are zero and which Exis are one? Yes, and that assignment is not really unique.

For example, in what I just said, the assignment of X one equal to zero and x two equal to zero corresponded to the Asus brand. So both the other brands were compared to this baseline in a sense. Therefore, we set Asus to be the baseline and x1 equal to 1 and x2 equal to 0 to indicate the Lenovo brand.

And x1 equal to 0 and x2 equal to 1 to indicate the Dell brand. But alternatively, we could have chosen x1 equal to 0 and x2 equal to 0 to represent the Lenovo brand. So make Lenovo the baseline. Yes, and then we could have chosen x1 equal to 1 and x2 equal to 0 to indicate the Asus brand, and had x1 equal to 0 and x2 equal to 1 to indicate the Dell brand.

In this case, the interpretation of the coefficients will be slightly different. They would reflect comparisons with respect to Lenovo that we choose to be the baseline. Yes, and beta 0 would correspond to the expected satisfaction score of Lenovo now, not Asus, and beta 1 would be interpreted as the difference between Asus and Lenovo, and beta 2 would be interpreted as the difference between Dell and Lenovo.

The important thing to remember is that when a categorical variable has k levels, it is k minus 1 dummy variables are required in the multiple linear regression analysis. Shall we do an example with more than one level of dummy variable? Yes. Should we do this with RAM or hard disk for the Basavaraju's data set? How about we try this on the household survey data that we had from Manjula Nair? Sure. Let's try using dummy variables on Manjula's data set for the household survey.

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Example of Categorical Variables – I

Let us recall our most recent attempt with Manjula Nayak's household survey data. We had tried a simple linear regression and two versions of multiple linear regressions. Our last attempt had a multiple linear regression model with the debt y as a function of three independent variables. Namely, the first income as x1, the monthly payment as x2, and utilities as x3.

Our regression model was the dependent variable y is equal to beta 0 plus beta 1 times first income plus beta 2 times monthly payment plus beta 3 times utilities plus the error epsilon. Tejas, what was our estimated regression equation in that case? B0 is minus 140378, B1 is 0. 017 for first income, B2 is 0. 96 for monthly payment, and B3 is 157. 95 for utilities. Yes, and what was the R square? What did we get? We got a value of 0. 708 from tool pack. Now with dummy variables, we are going to make it better. You seem determined. Alright, let us try. Recall that in the F test, we rejected the null hypothesis indicating that beta 1, beta 2, beta 3 are all equal to zero, and therefore confirming that a significant overall linear relationship exists between debt, first income, monthly payment, and utilities.

Also, the t test of significance for the individual coefficients also confirm the significance of all the three variables that we currently have in the model. So, now, the question is, what can we add to try and improve this model? We want to try a categorical variable. I would imagine that ownership would impact debt in a household.

Whether they own a house or not should impact their debt, right? Tejas, I like the way you think. As you can see, we have several choices of variables here. And so we could add any of those variables. But as a model builder, it is useful to add variables that we think will have a significant effect, at least intuitively.

That is always a good strategy, especially when we have several variables to add. But ownership has only one level. We want multiple levels. Yes, that is right. In the data set, we have zero to represent renting the house. And one to represent ownership of the house. But let us try Tejas intuition first.

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Ultimately, our goal is to find a model that explains the variation in debt better than the one that we have now. So our proposed regression model is that the dependent variable Y is equal to beta 0 plus beta 1 times first income plus beta 2 times monthly payment plus beta 3 times utilities plus beta 4 times x4 plus the error epsilon, where x4 takes a value of 1 if the household owns the house and 0 if they rent it.

Okay, here is the spreadsheet with first income, monthly payments, utilities and ownership as inputs to be given to the analysis tool pack and debt as the y variables. As you have done always, let us collect not only the numerical outputs, but also the standardized residuals and the plot of the residuals.

Let me wait until you have collected all of those outputs before we begin to discuss. Yes, I have all the outputs now.

Okay, so now let us discuss what we see in the model. What can you say about the R square in the model? The R square of the model has increased to 0. 712 from the 0. 708 that we had. I'm disappointed. Yes, so much for intuition, is it not? I told you, we should have tried something else. You will get your turn too, Ashwini.

But let us try to understand what happened. What happened to the standard error? The standard error in the model is now 18594. Yes, and that has reduced a bit from the earlier standard error of 18699. Let us now look at the outputs of the ANOVA analysis. The table shows that we can still reject the null hypothesis because the p value is very small.

Yes, and so this implies that we can reject the null hypothesis that beta 1, beta 2, beta 3 and beta 4 are all equal to 0. And a linear relationship therefore exists between debt, first income, monthly payment, utilities and ownership. Next, let us look at the table with the coefficients. What are the values of B zero B one, B2B three and B four B zero is minus 1 8 5 9 8 8.

B1 is 0. 017, B2 is 1. 05, and B3 is 200. Yes, and B4 is minus 12837. Does this mean that the household that owns has lower debt or 12837? Seems odd. Yes, they just, that is the correct interpretation of our coefficient B4. Other factors remaining the same, a household with ownership of their house has less debt.

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Perhaps the house is paid for, and so therefore, having that property allows them to live with lower debt. As we saw earlier, for a household that rents that property, we could get one estimated regression, which in this case would be B0 plus B1 times first income plus B2 times monthly payment plus B3 times utilities plus B4 times 0. And for a household that owns their property, we would get another estimated regression equation, which in this case, would be B0, plus B1 times first income, plus B2 times monthly payment, plus B3 times utility, plus B4 times 1. Well, B4 is minus 1 2 8 3 7. Yes, and that is the effect of ownership on debt.

Okay, what do the tests for individual significance say in this case? The individual t test for first income, monthly payment and utilities all look fine with low p values. Yes, and what about the significance of the new variable that we just added? The dummy variable corresponding to the ownership? The coefficient, as we saw, is minus 1 2 8 3 7.

The standard error is 5 0 0 3. To get the t statistic, we divide this coefficient minus 12837 divided by the standard error 5003, and we get a minus 2. 56. This is a t statistic with 495 degrees of freedom. So what is the p value? The associated p value is 0. 01, which is less than 0. 05. Yes, and that allows us to reject the null hypothesis that beta 4 is equal to 0.

So this implies that in the population ownership has a significant linear association with debt. Not happy, Tejas? Okay. It is a small win, but I expected R squared to do better. Okay. Well, what about the residuals? We should have four residual plots now, one for each independent variable. They all look equally distributed above and below the x axis.

Nothing to worry about. Yes. And the plot of the residuals against the predicted value y hat also looks very similar. And the standardized residuals? Same as what we have seen, some high values, but mostly okay. Yes. So Ashwini, do you still want to try and make this model better? Let us see what other variable could we include to try and improve the performance. So our current regression model is that the dependent variable y is equal to beta 0 plus beta 1 times first income plus beta 2 times monthly payment plus beta 3 times utilities plus beta 4 times x4 plus the error term epsilon, where x4 takes a value of 1 If the household owns the house and zero if they rent it.

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In our estimated regression equation, we got the coefficients of b0, b1, b2, and b4 to be equal to as follows. b0 was minus 185, 988. b1 was 0. 017. B two was 1.05 and B three was 200. We also got B four corresponding to the categorical variable to be minus 1 2 8 3 7. The R square of the model was 0.712. Let us consider location as another independent variable.

Okay, so now location is a categorical variable that can take four possible values of Southeast, southwest, northeast, and northwest. To include this in the model therefore, we will need to add four minus one. That is three dummy variables, but why would location impact debt? We don't know, but we will find out and at least we will learn how to add categorical variables with multiple levels in the model.

Let us define three new categorical variables, X five, x six, and X seven with the following interpretations. Let x5 be equal to 1 if the property is in the northwest, and 0 otherwise. Let x6 be equal to 1 if the property is in the northeast, and 0 otherwise. And let x7 be equal to 1 if the property is in the southwest, and 0 otherwise.

So what does x5, x6 and x7 all being equal to zero imply? It implies that the property is in South East. Yes, and that would be our baseline. Corresponding to x5, x6 and x7, we will have additional parameters beta 5, beta 6, and beta 7 in the model, in addition to the ones we already have. The corresponding coefficients in the estimated regression equation B5, B6, and B7 will tell us the effect of location on debt in comparison to the debt of a property in the southeast, our baseline location.

First, let us reorganize our data, so that we can use Analysis Toolpack. Here, I have the columns with all the X variables, first income, monthly payment, utilities, and then the categorical variables corresponding to ownership. Then, the three columns corresponding to categorical variables, Northwest, Northeast, and Southwest.

We also have the Y variable, the dependent variable corresponding to debt. Let us enter all of this into Toolpack and collect all the outputs. I will wait until you have it all done. Give me a minute. I'm almost done.

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Okay. Looks like you have it. So let us examine the outputs. Where shall we start? What is the R square for the new model? The R square of the model has increased to 0. 714 from 0. 712 that we had, with ownership added. No significant improvement, so disappointing. Yes, I agree. But you know, this has made Tejas feel better all of a sudden.

Let us try to understand what happened. What happened to the standard error? The standard error in the model is now 18589. Yes, and that has reduced a little bit from the earlier standard error of 18594. Barely better. Let us now look at the outputs of the ANOVA analysis. The table shows that we can still reject the null hypothesis because the p value is very small.

Yes, and this implies that we can reject the null hypothesis that all of beta 1, beta 2, beta 3, beta 4, beta 5, beta 6, and beta 7 are all equal to 0. And a linear relationship between debt, first income, monthly payment, utilities, ownership, and location is indeed significant. Next, let us look at the table with the coefficients.

What are the values of b0, b1, b2, b3, and b4? And what about our new coefficients, b5, b6, and b7? b0 is minus 184243, b1 is 0. 015, corresponding to first income, b2 is 0. 96 corresponding to monthly payment, and b3 is 200 corresponding to utilities. Yes, and these are all similar to what we got earlier. And what about B4 corresponding to ownership? Did that change much? It is minus 1 2 8 5 7.

Just some marginal change, just the same though, directionally. And what is B5 corresponding to Northwest? B5 is 4229. Does this mean that the household in the North West has more debt than the household in the South East? Yes, exactly. That is what the data says. That households in the North West have more debt than households in the South East.

The difference is on an average 4229. Then we have B6 corresponding to North East to be 2935 and B7 corresponding to South West to be 5211. Yes, and that is interesting. All other locations have higher debt than our baseline location of Southeast. Okay. Also, let us note that our definition of the categorical variables now gives us eight expected regression equations.

We have four equations based on the location, and for each location, depending upon whether B4 corresponding to ownership takes a value of 0 or 1, we will get 2 equations for each location.

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So that would be a total of 8 regression equations. So we could say we are building multiple models simultaneously.

Yes, in a sense, that is what we are doing. Ok, what does the t test for the individual significance say in this case? The individual t test for first income, monthly payment and utilities all look fine with low p values. What about the T-test corresponding to beta five or northwest? What is the coefficient b five, the standard error, the test statistic, and the P value? Corresponding to northwest, we have B five equal to 4229. The standard error is 3471 dividing B five by the standard error, we get the test statistic to be equal to 1.22. This is a T statistic. And the corresponding p value is 0. 22, which is greater than 0. 05. And this implies that we cannot reject the null hypothesis that beta 5 is equal to 0. So this test of significance is suggesting that we may be better off by not having this categorical variable in the model. Oh no, that's not good. Yes. What about the t test corresponding to beta 6 or northeast?

What is the coefficient b6, the standard error, the test statistic, and the p value? Corresponding to northeast, we have B6 equal to 2935, and the standard error is 2531. Dividing B6 by the standard error, we get the test statistic to be equal to 1. 16. This is also a t statistic, and the corresponding p value in this case comes out to be 0. 25, which is also greater than 0. 05. This implies that we cannot reject the null hypothesis that beta 6 is equal to 0. So this test of significance is also suggesting that we may be better off by not having this categorical variable in the model. Ashwini, another of your categorical variables kicked out.

What about the t test corresponding to beta 7, or southwest? What is the coefficient b7, the standard error, the test statistic, and the p value? Corresponding to southwest, we have b7 equal to 5211, and the standard error is given by 2881. Dividing B7 by the standard error, we get the test statistic to be equal to 1. 81. This is also a t statistic, and the corresponding p value is 0. 07, which is again, unfortunately, greater than 0. 05. This implies that we cannot reject the null hypothesis, that beta seven is equal to zero. So even this test of significance is suggesting that we may be better off not having this categorical variable

also in the model. Ashwini, sorry, the individual t tests are suggesting that you take out all the variables you wanted to put in. At least the ownership that I suggested was not kicked out.

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Okay. They just let us not get personal and Ashwini looks terribly disappointed. But this is teaching us a few good lessons.

Failures are the stepping stones to success, you know. So first, let us accept that when we started the model that had a good R squared of around 70%. First, let us accept that we started with a model that had a good R squared of around 70 percent before we added any of these categorical variables. So, it would be progressively harder to find a model that does better and better.

Even with ownership, we saw that the impact was not really that significant. But we did add some complexity to the model by including another independent variable corresponding to ownership. Now, with location, the model is encouraging us to not add more complexity to the model as the benefits are not worth it.

Building a good regression model requires striking a balance between complexity and performance. If you add too many variables, you could overfit the data and not have a good model. Just because we have collected several variables in our data, we should not be compelled to add them all into a model.

Regression is telling us which variables are relevant in explaining the variability in debt. Our so called failures to add categorical variables are essentially underlining the success that we had with the first three independent variables, namely first income, monthly payment, and utilities. So you are saying we did a pretty good job with those and so it is hard to improve it further that easily.

Yes, in a sense, that's what the regression model is telling us. And you are not just saying that to cheer us up. Well, that is what the model is telling us, is it not? Okay, if you say so. Well, maybe we should do another example with categorical variables. Perhaps that'll make us all feel much better.

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Example of Categorical Variables – II

Let us look at flat prices in Mysuru. Vijay Gowda, here we come. Let us revisit the model that we built using the data set we had from Vijay Gowda, who is a real estate analyst working for Magic Bricks. If you recall, he was analyzing the property prices for flats in Jayalakshmi Puram in Mysuru to determine the market rates for comparable properties.

His data set had 30 observations with variables such as selling price in lakhs, square feet area, number of bedrooms, number of bathrooms in the apartment, and also information about whether the property was located in a premium locality, gated complex, and had good amenities. We had developed a simple linear regression model with selling price as the dependent variable y and the area square feet as the independent variable.

We had estimated the regression equation to be 28. 37 plus 0. 0968 times x, the area in square feet. However, the R square was only 0. 44. So, we tried a multiple linear regression model to improve and explain more of the variability in the selling prices. Do you recall what happened? We added the number of bedrooms as an additional independent variable, x2, but the R squared did not improve.

Yes, and what was the reason that it did not improve? Multicollinearity. Area square feet and number of bedrooms had a significant positive correlation. Yes, exactly. And so, what did we do next? We decided to try the other quantitative variables in the data set, the number of bathrooms. And what happened when we did that?

Very similar problems. Again, multicollinearity effects. Yes. And all we were then left with were to try and improve the model where the qualitative variables. Qualitative variables that were corresponding to whether the property was located in a premium locality, a gated complex, or whether it had good amenities.

At that time, we said we will revisit that example once we know how to include qualitative variables into multiple linear regression. It is now time for us to do that. Are you ready, Tejas and Ashwini? Yes, let us try that. I'm hopeful. What about you, Ashwini? I'm not sure anymore.

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Okay, maybe you will all feel better and more confident if we try some improvements to this model.

Here is the data set. It is already provided to us with 0s and 1s data that could allow us to include categorical variables. The regression model that we currently have is y is equal to beta 0 plus beta 1 times x, where y is the selling price and x is the area in square feet. The estimated regression equation is y hat is equal to b0 plus b1 times x, where b0 is 28. 37 and b1 is 0. 0968. The R square for this model, as we saw is 0. 44. We have three choices of categorical variables that we can add to this model. The first is whether the property is located in a premium location or not. The data lists values of 1 if the property is indeed in a premium location, and 0 otherwise.

The second is whether the property is located in a gated community complex or not. The data lists values of 1 if the property is indeed in a gated community, and 0 otherwise. The third is whether the property has good amenities, swimming pool, fitness centers, etc or not. And the data lists values of 1 if the property has good amenities and 0 otherwise.

Which of these should we try first? You know, they say real estate is about location, location, and location. That is true. So let us try adding a categorical variable indicating if the property is in a premium location or not. So let us propose a regression model that is as follows. The dependent variable y is equal to beta 0 plus beta 1 times area square foot plus beta 2 times x2 plus the error epsilon.

Where x2 takes a value of 1 if the property is in a premium location and 0 if the property is not located in a premium location. Okay, here is the spreadsheet with the data. The selling price, the area square foot, premium location, gated communities, amenities, number of bedrooms, and number of bathrooms.

Let us get going with the analysis tool pack. Choose regression as the option. Let the y variable corresponding to the data for selling price, and let the x variables correspond to both area square footage and the premium location. So we skip number of bedrooms and number of bathrooms. Yes, let us skip those as we face multicollinearity issues when we added them.

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As we have done always, let us collect not only the numerical outputs, but also the standardized residuals and the plot of the residuals. Let me wait until you have collected all of those outputs before we begin to discuss. Do you have the outputs? Well, looks like you have them now.

Okay, let us look at the R square. What did you get? Huray, we get a value of 0. 66 from tool pack and that is a good improvement from 0. 44 that we had with just simple linear regression. Yes, there is an improvement. Wait before you celebrate, Ashwini. We don't know what the diagnostics will tell us. Yes, that is true.

Tejas has become more cautious now, after all these examples. So now, what does the output tell us about the adjusted R squared? The adjusted R squared is pretty close to the R squared, about 0. 63, and also better than the earlier value of 0. 43. Yes, we have been seeing that adjusted R square is being fairly close to the R square in all the models that we have seen so far.

Next, let us look at the standard error. We see that the standard error is 56. 35, which is far lower than the 70. 57 we got with just the simple linear regression model. Let us next examine the ANOVA output. The table shows that the sum of squares regression, SSR, is 165273, which is higher than the 111539 that we got earlier.

So adding the categorical variable seems to help us to explain more of the variability in the selling prices. The table also shows that the sum of squares error SSE is equal to 85728, which is lower than the previous value of 139462. And what is the sum of squares total or SST? The SST is 251001. As expected, SSR of 165273 is about 66 percent of the SST.

Yes, so the regression is explaining around 66 percent of the variation. So now let us look at the MSR and MSE. The MSR is 8 2 6 3 7, and the MSE is 3 1 7 5. And since we have two independent variables, the MSR is equal to the SSR divided by two and is equal to the 8 2 6 3 6 that we just saw. For the MSC, the degree of freedom is 27, and so we have the SSE divided by 27 to be equal to the MSC of 3 1 7 5.

Okay. So what does the F statistic come out to be then? The F statistic is the ratio of MSR over MSE, and that comes out to be 26. 03. Yes, and what does that mean in terms of our p value?

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It is close to zero. Yes, and so therefore, we conclude that this allows us to reject the null hypothesis that both beta 1 and beta 2 are equal to zero.

And that implies that there is enough evidence to claim that there is a significant overall linear relationship between selling price, area square foot, and premium location. Seems promising, is it not? Next, let us look at the table with the coefficients. What are the values of B0, B1, and B2? B0 is minus 3. 92. B1 is 0. 087 for area. And B2 is 86. 2. Yes. And how do we interpret this coefficient B1? For every square foot increase, the selling price increases by 0. 087 lakhs or 87, 000. And what is the outcome of the hypothesis test with the null hypothesis of beta 1 equal to zero. We need to look at the row corresponding to area square feet and it's coefficient B one.

Yes. And we look at the standard error, the test statistic, and the P value. We notice that the standard error is 0.016 and if you divide the coefficient B one, which is 0.087, by the standard error of 0.016, we get a value close to 5.31. And in this case, this is a t statistic with 27 degrees of freedom that has taken a value of 5. 31. For this t statistic, what is the p value? It is close to zero. And so, what does that imply? It implies that we can reject the null hypothesis. Yes, confirming that a linear relationship between selling price y and area square foot x1 is significant. Okay. Now on to our categorical variable X two and its coefficient B two.

How do we interpret that? If X two is equal to one, that is, if the property is indeed in a premium location, then the selling price is higher by B two, which is 86.2 lakhs. Oh, location really makes a difference. Yes. And what is the outcome of the hypothesis test? When the null hypothesis of beta 2 is equal to 0, we need to look at the rho corresponding to premium location and its coefficient, B2.

And we look at the standard error, the test statistic, and the p value. We notice that the standard error is 20. 95. And if we divide the coefficient B2, which is 86. 2 by the standard error of 20. 95, we get a value that is fairly close to 4. 11. So in this case, this is a t statistic with 27 degrees of freedom that is taking a value of 4. 11. And for this t statistic, the p value is close to zero. And what does that imply? It implies that we can reject the null hypothesis and confirm that the linear relationship between selling price Y and premium location X2 is significant. Yes.

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Let us now look at the residuals. We have the residuals as well as the predicted selling price given by our estimated regression equation.

What can we say about the standardized residuals? It is better now. None are above 3 in absolute value. Yes, that was not the case before. So adding the categorical variable has helped us to explain more of the variability. And what does the plot of our residuals against x1, which is the selling price, telling us?

The plot of residuals against x1 is random, both positive and negative values. And what does the plot of the residuals against x2, which is premium location, telling us? That plot of residuals against X two is also random, also looks fine. And what about the plot of residuals against y hat, the predicted values of the selling price? That is also equally distributed above and below the x axis. Yes. Again, nothing to be concerned about. And what about the plot of standardized residuals against Y hat, the predicted selling price? Similar insights like what we saw when we looked at the table of standardized residuals. Yes, so we see that adding this categorical variable corresponding to premium location really gave us a much better model.

We saw that area square feet was an important variable, however the other quantitative variables number of bedrooms and number of bathrooms were not required as they were highly correlated to the area square foot that was already present in the model. But adding this categorical variable corresponding to location helped us to better explain the variation of price in the data.

So should we try adding some of the other categorical variables into the model like amenities and whether the property is in a gated society? Yes, we should try that. And perhaps that will help us explain more of the variability in the selling prices. If the regression diagnostic indicates otherwise, we can rethink.

But let us try that. So let us propose a regression model that is as follows. The dependent variable y is equal to beta 0 plus beta 1 times area square feet plus beta 2 times x2 plus beta 3 times x3 plus beta 4 times x4 plus the error epsilon. Where, as before, x2 takes a value of 1 if the property is in a premium location and 0 if it is not located in a premium location.

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And the new variable is x3, takes a value of one, if the property has amenities and zero if it does not. And X four takes a value of one if the property is in a gated community, and zero if it is not. Okay, so here is a spreadsheet with the data. The selling price, the area square feet, premium location, gated community, amenities, number of bedrooms, and number of bathrooms.

Let us get going with analysis tool pack. Choose regression as the option. Let the y variable corresponding to the data of selling price, and let the x variables corresponding to both area square feet, premium location, amenities, and community. As we have done always, let us not only collect the numerical outputs but also the standardized residuals and the plot of these residuals. Let me know when you have all of these results.

It looks like we have all the outputs. Okay. So as always, let's start by looking at the R squared. What did you get? We got a value of 0. 80 from tool pack. That is a good value. We might have a good model. Yes, that is a good improvement from the 0. 66, which was already a good improvement from the 0. 44 that we had earlier with only area square foot in the model.

Looks like more of the variability in flat prices were because of the categorical variables than just the quantitative variables. Yes, that is true and it is a point to keep in mind when we collect data for decision making. It is not that only quantitative data is actually useful. The qualitative data we might collect may be very relevant.

And multiple linear regression has a very nice way of having a blend of both qualitative and quantitative variables in the model. Okay. Now, what does the output tell us about the adjusted R squared? The adjusted R squared is 0. 77, also better than the 0. 63 that we had earlier. Next, let us look at the standard error.

And we see that has improved to 44. 64. That is also lower than 56. 35 with only one categorical variable. Adding multiple categorical variables did not help with Manjula's household survey, but here it seems to have helped. Let us look at the detailed diagnostics before we pass any conclusions.

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Let us examine the ANOVA output. The table shows that SSR is 201194, which is a good jump and much higher than the 165273 we got earlier. Yes, and adding the four categorical variables seems to help us explain more of the variability in the selling prices. The table also shows that SSE to be equal to 49808, which is lower than the previous value.

And what about the SST? Just to verify, it is still 251001, as it should be. And so therefore, our SSR of 201194 is about 80 percent of the SST. Yes, and the regression is explaining close to 80 percent of the variation. Let us now look at MSR and MSE. MSR is 5 0 2 9 8 and the MSC is 1 9 9 2. Since you have four independent variables, the MSR is equal to the SSR divided by four and is equal to 5 0 2 9 8.

For the MSC, the degree of freedom is 25, and so we have SSC divided by 27 to be equal to 1 9 9 2. Okay, and what does the F statistic come out to be? The F statistic is the ratio of MSR over MSE, and that comes out to be 25. 25. Yes, and what does that mean in terms of our p value? It is close to zero. This allows us to reject the null hypothesis that all beta 1, beta 2, beta 3, and beta 4 are equal to zero.

And that implies that there is enough evidence to claim that there is a significant overall linear relationship between selling price, area square foot, premium location, amenities, and community. Next, let us look at the table with the coefficients. What are B0, B1, B2, B3, and B4? B0 is minus 43. 99. B1 is 0. 078 for area. B2 is 95. 66 for premium locality. B3 is 35. 74 for amenities. And B4 is 55. 38 for community. Yes. And what is then our estimated regression equation? It would be that the estimated selling price, y hat, is equal to 44. 99 plus 0. 078 times x1, plus 95. 66 times x2 plus 35. 74 times x3 plus 55. 38 times x4.

Actually, with three categorical variables, we have now eight estimated regression equations embedded in our model, two for each categorical variable. How do you interpret the coefficient b1? It is reduced from 0. 087. to 0. 078. For every square feet increase, the selling price increases by 7800. Yes, it has reduced.

Right. So, is this an indication of omitted variable bias that you talked about? Yes, B1 was trying to account for the categorical variables that we had omitted. And now, it is perhaps closer

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to what it should be than it was before. What is the outcome of our hypothesis test with the null hypothesis of beta1 equal to 0?

We look at the row corresponding to area square foot and its coefficient, b1. We look at the standard error, the test statistic, and the p value. We notice that the p value is very small, which implies that we can reject the null hypothesis. Confirming a linear relationship between selling price, y, and the area square foot, x1, is indeed significant.

Okay, now, on to the categorical variable, x2 and it's coefficient of B two. How do we interpret that? With the other independent variables being kept constant, if X two is equal to one, then if the property is a premium location, we say that the selling price is higher by B two, which is 95.66 lakhs. And what is the outcome of the hypothesis test with the null hypothesis of B two being equal to zero?

We need to look at the row corresponding to premium location and its coefficient b2. We look at the standard error, the test statistic, and the p value. Again, we see that the p value is close to zero. And what does that imply? It implies that we can reject the null hypothesis and confirm that the linear relationship between selling price y and premium location x2 is significant.

Yes, indeed. Now, on to the new categorical variables that we added, the categorical variable x3 and its coefficient b3. How do we interpret that one? With the other independent variables being kept constant, if x3 is equal to 1, that is, if a property has amenities, then the selling price is higher by B3, which is 35. 74 lakhs. What is the outcome of a hypothesis test with the null hypothesis of beta 3 equal to 0? We need to look at the row corresponding to amenities and its coefficient B3. And if you look at the standard error, the test statistic and the p value, and we see that the p value is 0. 04. And what does that imply?

It implies that we can reject the null hypothesis and confirm that the linear relationship between selling price y and amenities x3 is indeed significant. Next, we see the third categorical variable, the categorical variable x4 and its coefficient b4. How do we interpret this coefficient? With the other independent variables being kept constant, if x4 is equal to 1, that is, if a property is in a gated community, then the selling price is higher by B4, which is 55. 34 lakhs.

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What is the outcome of the hypothesis test with the null hypothesis of beta 4 being equal to 0? Again, we look at the row corresponding to the community and its coefficient B4. We look at the standard error, the test statistic, and the p value. And the p value in this case is 0. 003. That implies that we can reject the null hypothesis and confirm that the linear relationship between selling price y and community x4 is indeed significant.

That is nice. All the categorical variables can be retained in the model. Yes, and unlike what happened with Manjunath Nayak's data set, where the p values raised red flags, here, the p values are telling us that all these categorical variables are indeed significant. So, the lesson here is that only variables that matter, qualitative or quantitative, need to be kept in the model.

It is also telling us how some of these categorical variables by themselves impact the selling price. Yes, that is true. Such a model could be really helpful for both a buyer and a seller. Yes, that's right. Shall we look at the residuals and confirm if all is okay? Let us now look at the residuals. We have the residuals as well as the predicted selling price given by our estimated regression equation.

What can we say about the standardized residuals? Again, they look okay. None are above 3 in absolute value. Yes, and the categorical variables have helped us explain more of this variability. And what about the plot of the residuals against x1, x2, x3, and x4? The plot of residuals against x1, x2, x3, and x4 are all random, both positive and negative values.

Yes, that is true. Except for the categorical variables, the observations are bunched around their values of 0 and 1. Yes, expected to a certain extent. And as long as there are positive and negative values nicely distributed, it is okay. There is no bias. And what about the plot of residuals against Y hat, the predicted value of the selling price?

That is also equally distributed above and below the x axis. Yes. Again, nothing to be concerned about. And what about the plot of the standardized residuals against Y hat, the predicted selling price? Similar insights like what we saw when we looked at the table of standardized residuals. Yes, and so we see that adding the three categorical variables corresponding to premium location, amenities and community really gave us a much improved model.

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And hopefully, Tejas and Ashwini, in the process, you are learning a little bit about model building. We are adding variables stepwise to progressively improve the model. We could also just put all the variables we have in the first group, then delete variables one by one, right? Yes, of course, you could do that too.

But I think the approach that we are following is better because it is teaching us something and providing us with insights as we go along. Okay, so that is enough for now about the inclusion of categorical variables in multiple linear regression. Let us now talk about another important concept known as interaction variables.

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Interaction Variables

In multiple linear regression equations, sometimes we may want to introduce interaction variables. What are interaction variables? Interaction variables are a product of two independent variables, usually a product of a continuous variable and a categorical variable. Including the interaction variables in a regression model, many times helps us to check for the existence of some conditional relationships between a dependent variable and the independent variables.

For example, consider the sample data obtained from Jyoti Hegde's survey from IIIT, Dharwad that we saw in module one. Here is a sample of that data. In the data set, we have data related to salary, gender, men or women, and work experience in months for a sample of 30 respondents. Let us assume that salary is a dependent variable denoted by y. Let gender be the first dependent variable x1 and a categorical variable. Next, let work experience be the second independent variable, x2, a continuous variable.

Now, let us define the interaction between gender and work experience as the product of x1 and x2. Then, our regression model is given by y is equal to beta 0 plus beta 1 times x1 plus beta 2 times x2 plus beta 3 times x1 times x2 plus epsilon, the error term. In this regression equation, the term x1 times x2 is an interaction variable.

Wait a minute. Does this not make it a non linear regression model? Why? Because we are multiplying x1 and x2. Think about that for a minute, Ashwini. Can you tell me what do we mean when we say that a regression model is linear? What should it be linear in? Oh sorry, it should be linear in coefficients beta 0, beta 1, beta 2, etc.

So in this case, even though we are multiplying x1 and x2, it will still be a linear regression model. Yes, exactly. Alright, let us move on. Let us assume that women are coded as 1 in the gender variable and men are coded as 0 in the gender variable x1. Then, corresponding to x1 equal to 1 or women in the data, we we have the following regression model.

The dependent variable y is equal to beta 0 plus beta 1 plus beta 2 plus beta 3 times x2 plus epsilon. For x1 equal to 0, corresponding to men in the data, we have the following regression

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model. The dependent variable y is equal to beta 0 plus beta 2 times x2 plus epsilon. If the regression coefficients beta 2 and beta 3 are statistically significant, then the marginal change in the value of y for a unit change in x2 will be different for men and for women.

That is, the marginal change in Y for women as X2, as work experience changes, will be given by the sum of beta2 plus beta3. However, the marginal change in salary for men when work experience X2 changes would be given by just beta2. In other words, the relationship between the dependent variable y and the work experience x2 is conditioned on the variable x1 that denotes the gender.

So, this regression model will result in two different equations for men and women as we just saw. Yes, Ashwini, and it will lead to different models for each gender. So, in this case, the variable x1 corresponding to gender is known as the moderation variable. A variable is called a moderation variable when it affects the relationship between a dependent variable y and another variable x2.

In the case of our example, it corresponds to work experience. In other words, the regression coefficient value for work experience x2 depends on the other variable x1 corresponding to gender in our example. Let us now use Analysis Toolpack to run the regression equation for Jyothi's sample data and see what we observe.

Most of the analysis follows through in the same way as we have done for all the examples so far on multiple linear regression. We have our basic regression model to be given by y is equal to beta 0 plus beta 1 times x1, plus beta 2 times x2, plus beta 3 times x1 times x2, plus epsilon, the error term.

In the analysis tool pack, we select the data corresponding to salary to be the dependent variable, y, and then select the input variables, x, according to gender, work experience, and the product of gender and work experience. As we have always done, we select the numerical outputs and all the outputs related to the residuals as well.

The output analysis from the Analysis Toolpack is given here. Let us look at the R square value. What is it? The R square is pretty good, 0. 92. Yes, and the adjusted R square is 0. 91, also very

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good. And what does our F test for overall significance tell us? The p value is small, and so we can reject the null hypothesis that beta1 equals to beta2 equals to beta3 equals to 0.

Yes, and what about the p values for the t test? The t test for significance of the individual parameters? The p values are all below 0. 05, and so all the independent variables are significant. Good, and so then what is our estimated regression equation? So the estimated regression equation is given by Y hat is equal to 13450.2 minus 7771. 68 times the gender x1 plus 293. 8 times the work experience x2 minus 241. 6 times gender multiplied by work experience x1 times x2.

Then the specific equation for women corresponding to x1 equal to 1 can be written as as y is equal to 13450. 2 minus 7771. 7 plus 293. 8 minus 241. 6 times x2 or work experience or equivalently as y is equal to 5678. 5 plus 52. 2 times x2 or work experience. And the specific equation for men corresponding to x1 equal to 0 can be written as y is equal to 13450. 2 plus 293. 8 times the work experience x2. That is, the change in salary for women when work experience increases by one month is 52. 2 whereas for men the change is 293. 8.

In other words, the salary for men increases at a higher rate than that compared to women with work experience. So, you can see how interaction variables can be used to make important inferences in regression models. Any questions on the interaction variables? No questions, but I'm impressed by the overall versatility of linear regression models.

It really can capture many things. Yes, it does. And we have seen only a limited set of features in our short time in these modules. Yes, I am quite impressed too. And I think we should not forget that regression equation is essentially making inferences on the population. Yes, let us not forget that. So all our validated models are allowing us to use the information from one sample to infer about the linear relationships, whether they are going to be significant or not in the overall population. Okay. So we will leave our discussion on the interaction variables at that level for now. That also brings us to the end of the two main topics we wanted to cover in this module.

Our main goal in this module was to learn about multiple linear regression with categorical variables. Although, like in the previous module, we are still focused on multiple linear

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regression models, and we try to establish a linear relationship between a dependent variable y and several independent variables x. Here, some of these independent variables can be qualitative or categorical variables. The other variables could be quantitative, like we had in the previous module.

We saw the basic multiple regression model and the associated equation, and the estimated multiple regression equation. As we did in the previous module, the least squares technique was used to determine the coefficients in a multiple regression equation. We evaluate the model based on the multiple coefficients of determination, and like we did in the previous module, we conducted the test for overall significance, namely the F test, and the individual significance, namely the T test.

And we analyzed the residuals too. And of course, we applied them to some examples. And from an analysis perspective, the Excel perspective, it did not change much of what we needed to do. Yes, that is true. Apart from setting up your data in the required way, no other change was required. And we learned some bitter lessons.

Sometimes adding categorical variables do not bring you much benefit. Yes, indeed. Just because you have the data, it doesn't mean that adding more and more variables will help you build a better model. Finally, we saw how to define interaction variables to help model and understand important interaction effects.

And all of this leaves us impressed with what regression can do. Yes, that is true. The technique is very powerful. And not to forget how easy it is to do when you have a tool like Excel. Yes, of course. It takes the effort away from the computation and focuses it on the interpretation of the outputs, which is essential for managerial decision making.

Okay. Time for us to wrap the course now with a quick summary and review of what we have covered in all the modules.

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Wrap up

When you started this course, you probably wondered why you would need a course on Business Statistics for Entrepreneurs. All business managers, whether you are an entrepreneur yourself or working for an entrepreneur, will need to make decisions. And managers are expected to make decisions based on data and facts.

The process of decision making itself can be complex. In many cases, business decisions are made based on intuition and experience. Yes, Tejas, but in many cases, that can be very risky. It is always better to back up your decisions with analysis based on data. So therefore, it is important to know how to summarize the available data, how to analyze and interpret the data.

And how to make inferences from the data that are relevant to your decisions. And also how to quantify the risk associated with the decisions due to the limitations of available data. These skills have always been important, but they are more important today than ever before. Yes, because of the availability of vast amount of data.

Absolutely and the availability of a variety of tools and techniques that support data driven decision making. Ignoring the potential of statistics and data driven decision making essentially leaves you and your business vulnerable to the competition. Providing you exposure, experience, and confidence in the use and application of various statistical methods was one of the main purposes of this course.

In the first module, we looked at various types of data, both qualitative data and quantitative data, and the various ways in which they could be measured and reported. We looked at the various ways of visualizing data. These techniques could be very helpful in summarizing the collected data. Then, we looked at numerical summaries of data, both for measures of central tendency, like the mean, median, mode, and so on, and measures of dispersion, like the range, the standard deviation, and quartiles.

We also covered various basic concepts in probability. All of this provided the foundation for subsequent modules. In Modules 2 and 3, we covered random variables and probability distributions. We saw that random variables can be discrete or continuous. For discrete random

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variables, the probability mass function is a function that connects the values that the random variable can take to the associated probability.

Using the probability mass function, we compute useful measures such as expectation, variance, the cumulative probability, and the tail probability. We also show how to use the probability mass function to calculate the inverse calculations. Yes, that is right. And what types of discrete random variables did we see?

We saw discrete random variables such as the Bernoulli random variable, the binomial random variable and the Poisson random variables. We saw several applications of these random variables too. And how to use the template to do the calculations fairly quickly. Yes, the main part is to convert the business question into a relevant probability question.

Once we do that, the calculations are relatively easy. Then we looked at continuous random variables. Here we have the probability density function that essentially plays the same role as the probability mass function. Using the probability density function, we can calculate the expectation, the variance, the cumulative probability, and the tail probabilities.

Here too, we saw how to do the inverse calculations. Yes, and we saw continuous random variables such as the uniform distribution, the exponential distribution, and of course, the normal and the standard normal distributions. What else did we do, Tejas? Then we looked at properties of linear combinations of random variables and its applications.

Yes, and in particular, we looked at linear combinations of normal random variables. Then we saw the relatives of the normal random variables, the chi square distribution, t distribution, and f distribution. Yes, and all of which came handy in subsequent modules. Although the names are different, they all have essentially the same characteristics, and we are interested in very specific properties of these distributions.

Then we made an important step and leap in module 4 from the sample data to statistical inference. For this, we had to recognize that all sample statistics, sample mean, sample variance, and sample proportion are essentially random variables. And do you remember why

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that is the case, Tejas? Because the sample itself is collected in a random manner from the population.

Yes. And the distribution from which we obtain estimates of these quantities when we collect sample data from a population is known as its sampling distribution. We saw the sample statistics for different population parameters. Then we discussed that the analysis of data is usually motivated by the need to make a decision that affects not only just the sample, but also the population at large.

And so we saw how the sample statistics and the point estimators could be used to infer useful properties about the population parameters. To achieve this, we use the help of a powerful and important result in statistics. Do you remember what that was, Ashwini? It was the central limit theorem. Yes, the central limit theorem helped us to leap from sample statistics to population parameters.

And we saw that although we cannot precisely say what these population parameters are going to be, from just a small sample that is drawn randomly from a population. We can, nevertheless, provide very specific probability related statements about where the population parameters are likely to be, based on just a single sample that we have extracted from the population.

We saw the specific applications in the case of X bar, P bar, and S. Then, in Module 5, we used the sample data to make inferences about the population. We did that through two important techniques. The first was confidence intervals, and the second was hypothesis tests. With respect to confidence intervals, we were able to use the point estimates obtained from the sample and provide interval ranges within which the population mean mu or the population proportion p or the population variance sigma square are likely to be.

For the case of population mean and proportion, these intervals were written as a sample point estimate, plus or minus a margin of error. Yes, that is correct. And in the case of population variance, what did we do? We were directly able to write the population variance to lie within a lower limit and upper limit with a certain level of confidence.

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Yes, exactly. And then we looked at hypothesis testing, which also allows us to make claims about population parameters. Claims such as whether a population parameter is equal to a certain value, greater than a certain value, or less than a certain value. We did this for population mean mu, population proportion p, and population variance sigma square.

In doing a hypothesis test, we need to be careful about how to structure the test, particularly when writing down the null hypothesis and the alternate hypothesis. It always helps to see who is trying to make the claim about the population parameter. Yes, that is the first thing to keep in mind. And the second thing to keep in mind is what is the specific claim that they are trying to make.

Once a hypothesis test is set up and the significance value is determined, then we assume that the null hypothesis is true at equality and compute the test statistic. We note that the random variable that the test statistic has sometimes corresponds to a t distribution, a standard normal distribution, and sometimes it has a chi square distribution.

Then we compute the p value, the probability of making a type 1 error. Yes, and if the probability of making a type 1 error is small enough, then we are able to reject the null hypothesis. Otherwise, we do not reject the null hypothesis. Both confidence intervals and hypothesis tests have a lot of applications in making statistical inferences from a single random sample obtained from the population.

In Module 6, we continued our journey in statistical inference by looking at another important technique known as regression. In particular, we started with simple linear regression. In simple linear regression, the regression model is y is equal to beta 0 plus beta 1 times x plus epsilon. Then we used sample data and the least squares method to develop the estimated regression equation y hat is equals to b0 plus b1 times x.

Yes, and then we wanted to validate our model. And to do that we define the concept of coefficient of determination as a measure of the goodness of fit of our model. Yes, the coefficient of determination is a ratio that tells us the proportion of the variation in the dependent variable y that is explained by our estimated regression equation.

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We saw that in this process of validation, the assumptions behind the linear regression, and especially the assumptions about our error term epsilon, are critical for the t test and the f test that we do in our validation exercise. These tests then told us whether the relationship between the two variables, y and x,

are indeed significant. Then we use Analysis Toolpack to conduct the regression analysis for several examples using Excel. We also saw how to conduct residual analysis to validate the assumptions, identify outliers, and influential observations. In Module 7, we expanded the ideas behind simple linear regression to multiple linear regression, where the dependent variable y depended on more than one independent variable. We saw the basic multiple regression model to be equal to y is equal to beta 0 plus beta 1 times x1 plus beta 2 times x2 and so on up to beta p times xp plus epsilon. The expected value of y or E of y was then equal to beta 0 plus beta 1 times x1 plus beta 2 times x2 and so on up to beta p times xp.

We then use sample data and least square method to develop the estimated regression equation. y hat is equal to b0 plus b1 times x1 plus b2 times x2 and so on up to bp times xp. We saw that these coefficients b0, b1, b2 and so on up to bp are sample statistics. Yes, and they are used to estimate the unknown model parameters beta 0, beta 1 and so on up to beta p in the population.

We saw a generalized version of the coefficient of determination known as the multiple coefficient of determination. Yes, and we also discussed the assumptions behind the error term epsilon in this model. We then did the F test for overall significance of the linear relationship and the T test for the parameters beta 1, beta 2, and so on up to beta p to test for the significance of the individual parameters in the model. We also saw concepts of multicollinearity and omitted variable bias. We analyzed several examples using analysis tool pack, and in the process of comparing and improving the models, we learned about model building.

Yes, it is indeed and iterative process. We also saw how we can analyze residuals to detect outliers and influential observations. Yes, and in this final Module 8, our main goal was to learn about multiple linear regression with categorical variables. Like in Module 7, we were still focused on multiple linear regression models, and we tried to establish a linear relationship between a dependent variable y and several independent variables x.

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But here, some of these independent variables can be qualitative or categorical variables, and the other variables could be quantitative, like we had in the previous modules. As we did in the previous modules, the least squares technique was used to determine the coefficients in a multiple linear regression equation.

We evaluated the model based on the multiple coefficient of determination like we did before. And like we did in the previous module, we conducted the test for overall significance, namely the F test, and the individual significance, namely the T test. We analyzed residuals too, and of course, we applied them to examples.

We saw that apart from setting up your data in the required way, not much change was needed in the regression analysis procedures. And we learned that sometimes adding categorical variables do not bring you much benefit. Yes, indeed. Just because you have the data, it does not mean that adding more and more variables will help you And finally, we saw how to define interaction variables to help model and understand important interaction effects.

As you can see, we have covered quite a bit of content in the eight modules. Some final words before we close. No doubt there is theory, and there is a fair amount of theoretical rigor behind the different concepts that we have covered in this course. But we made an effort to focus less on the theoretical derivations and intricacies.

Because of this intent, some purists would argue that we were careless in our choice of words or terminology at times. But we believe a focus on the applications and the interpretation of concepts in a variety of business settings is more important. We believe the theory behind any subject is better appreciated when you see its applications.

Hopefully, at the end of this course, you have a good understanding of the various statistical tools and techniques, and more importantly, a better awareness and confidence in the use of these techniques in a variety of business settings. I encourage you to apply these tools and techniques to solve real world decision problems that you will face in the future.

Like with any skill, you get better with practice. Merely watching videos of great sportspersons does not make you a great player. You need to learn from the videos and then step on the field

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and apply your learnings. If you falter, come back, review, adjust, and improve. I wish you the very best in becoming a successful entrepreneur.

Now, before we close, a time for some thank yous. First and foremost, thank you for registering for the program and giving us an opportunity to support your entrepreneurship ambitions. Although I might be the main face you see on the camera in each module, there are a lot of people that have spent countless hours behind the scenes trying to enhance the quality of your learning experience.

It would be unfair to wrap up this course without acknowledging their effort. The leadership and administration at both IIMB and IIMBx deserve acknowledgement for their vision and support. Several faculty have been involved in designing and curating this curriculum. Several staff at IIMBx and MOOCs and other functions at IIMB have been involved in the planning, the production, and the execution.

The camera team, the subject matter experts, the program research assistants, the editing team, and staff involved in providing IT and LMS support. A huge thank you to all of them. And finally, a personal thanks to Tejas and Ashwini. Without you, I could not have done this. Thank you all once again, and best wishes from all of us here at IIMBx.

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